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Overview

- 1 Motivation
- 2 Approach
- 3 Experiments
- 4 Results
- 5 Discussion
- 6 Conclusion

Goals and Research Question

Base Agent

- combination of Behavioral Cloning (BC) and Proximal Policy Optimization (PPO)
- strong baseline, but weaknesses against several opponents

Goal

- improve the Base Agent with League Training (LT)
- training against a more diverse set of opponents
- focus on robustness and generalization

Research Question

- How much does LT with PPO improve robustness and generalization of the BC/PPO-pretrained agent?
- How strongly does this depend on the diversity of initial agents and BC experts?

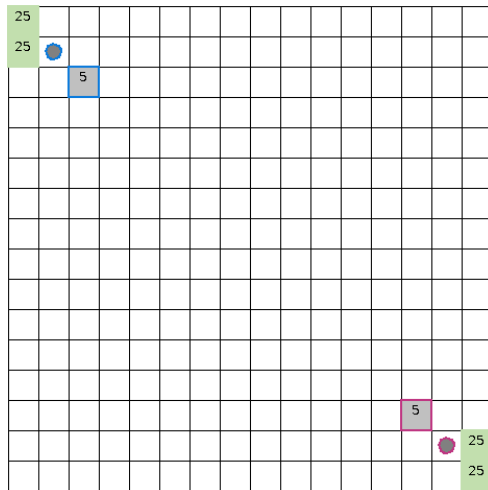
Setting

MicroRTS

- simplified RTS environment for RL research
- still contains resource management, unit production and combat
- supports controlled and repeatable experiments

Main RL challenges

- large combinatorial action space
- simultaneous decisions
- delayed and sparse rewards
- robustness against diverse opponents



Click the play area in Foxit Reader to start the embedded MicroRTS example video.

Training Pipeline



BC and BC-FT

- imitate experts
- BC-FT: only winning replays

PPO

- iteratively update policy
- refine the policy through exploration

LT

- train against an evolving opponent population
- add exploiters

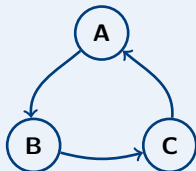
PPO fine-tuning

- repair the main remaining weakness after LT
- add WorkerRushAI as opponent

From SP to LT



cyclical learning of SP



directions the learning agent trains towards

Why LT helps

- improves stability by identifying and exploiting weaknesses
- creates diverse set of opponents without the main agent having to find them through exploration

LT Design

Main Agent

- primary policy
- resulting agent

Historical Agents

- frozen checkpoints kept in the opponent pool
- created by all other league agent types

Main Exploiters

- exploit weaknesses of the Main Agents
- resets, when checkpointing

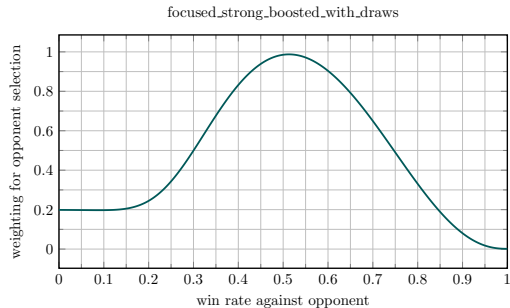
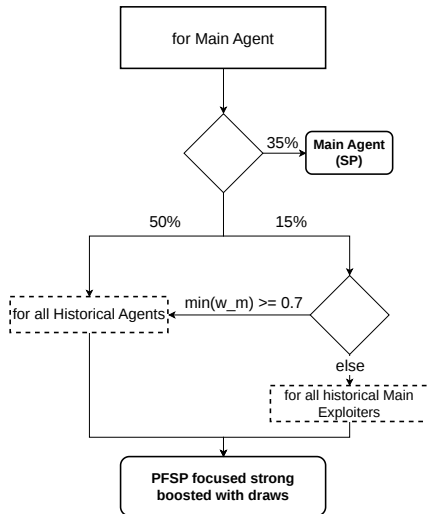
League Exploiters

- exploit broader weaknesses across the league
- has a chance of resetting when checkpointing

Key idea

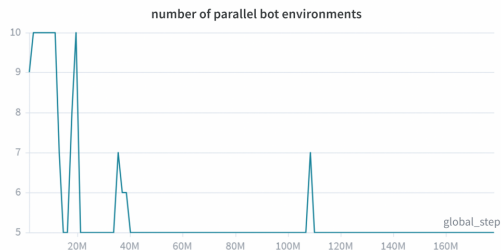
LT turns the robustness against diverse opponents into part of the optimization target.

LT Main Agent Matchmaking

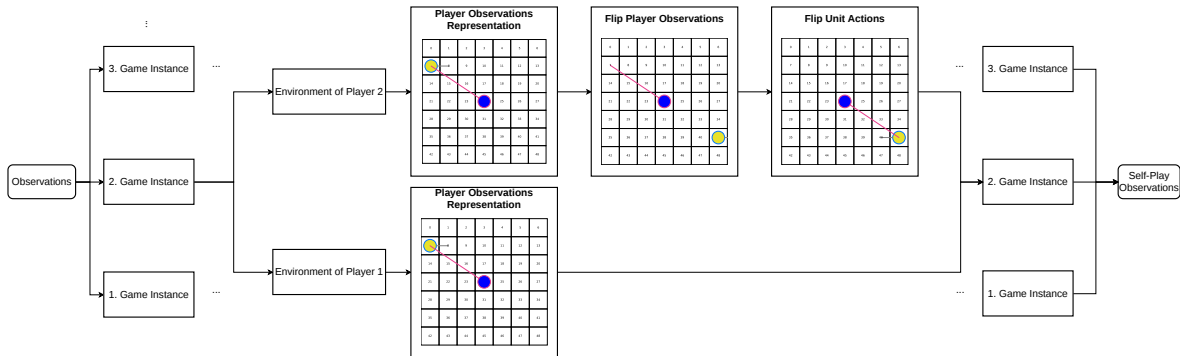


Bot Environments

- additional dynamic bot environments (CoacAI and Mayari)
- scripted bots: CoacAI, Mayari, WorkerRushAI, PassiveAI, LightRushAI, RandomAI and RandomBiasedAI
- MCTS based bots: Rojo, MixedBot, Izanagi, Tiamat, Droplet, GuidedRojoA3N and NaiveMCTS



Observation Adjustment for SP



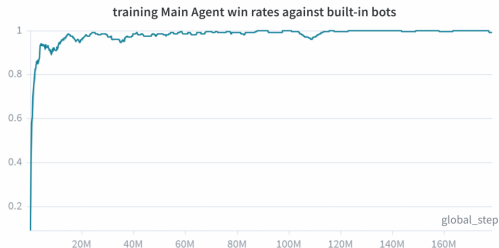
Player 2 observations are transformed so SP remains compatible with a Base Agent only trained as Player 1.

Other Implementation Changes

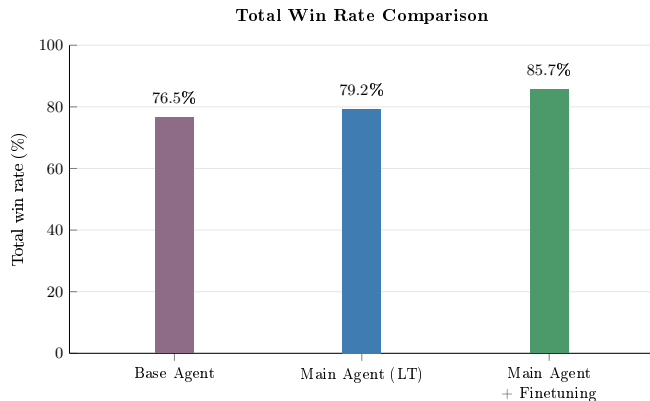
- delta score
- annealed entropy coefficient
- new initial agents

Experiment Setup

- map: basesWorkers16x16A
- LT setup: 1 Main Agent, 4 Main Exploiters, 1 League Exploiter
- 25 environments per learning agent
- 5 to 10 additional dynamic bot environments
- approximately 130 million LT steps
- training time: about 100 hours
- maximum setup size: 160 parallel environments
- evaluations over at least 100 games against 12 opponents after LT and after fine-tuning



Main Quantitative Result



- overall loss rate drops from **7.7%** to **2.2%**

Evaluation Results

Opponent	Main Agent without new initial agents	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
<i>bots used in training</i>				
CoacAI	100.0	96.0	100.0	100.0
Mayari	100.0	67.3	100.0	100.0
<i>Built-in bots</i>				
WorkerRushAI	31.0	100.0	35.0	99.5
PassiveAI	100.0	99.7	100.0	100.0
LightRushAI	100.0	98.0	100.0	100.0
RandomAI	100.0	99.0	100.0	100.0
RandomBiasedAI	91.0	79.0	97.0	97.5
<i>Other bots</i>				
Rojo	100.0	81.0	100.0	100.0
MixedBot	45.0	43.3	57.0	60.5
Izanagi	68.0	71.7	70.0	72.5
Tiamat	94.0	93.0	99.0	99.0
Droplet	45.0	67.7	62.0	75.5
GuidedRojoA3N	64.0	74.7	87.0	88.5
NaiveMCTS	0.0	0.7	2.0	6.5
overall	74.1	76.5	79.2	85.7

Evaluation Results

Opponent	Main Agent without new initial agents	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
<i>bots used in training</i>				
CoacAI	0.0	2.0	0.0	0.0
Mayari	0.0	27.3	0.0	0.0
<i>Built-in bots</i>				
WorkerRushAI	63.0	0.0	55.0	0.5
PassiveAI	0.0	0.0	0.0	0.0
LightRushAI	0.0	2.0	0.0	0.0
RandomAI	0.0	0.0	0.0	0.0
RandomBiasedAI	0.0	0.0	0.0	0.0
<i>Other bots</i>				
Rojo	0.0	5.3	0.0	0.0
MixedBot	33.0	34.3	20.0	13.5
Izanagi	15.0	16.3	7.0	6.0
Tiamat	6.0	7.0	1.0	0.5
Droplet	22.0	9.0	22.0	10.5
GuidedRojoA3N	0.0	1.7	0.0	0.0
NaiveMCTS	0.0	2.7	0.0	0.0
overall	9.9	7.7	7.5	2.2

Interpretation of the Results

hypothesis:

- LT turns robustness against diverse opponents into part of the optimization target
- LT helps to find a better local optimum that is then further improved by fine-tuning, while pure PPO often gets stuck in not as generalized local optima.

Player 1 Priority

Observation

MicroRTS gives player 1 action priority, so a potential concern is that self-play style training could accidentally favor player-1-specific behavior.

Result from the thesis

In one self-match evaluation of the same agent over 200 games, player 1 achieved 38% wins, 28% draws and player 2 achieved 34% wins.

- minimal effect on evaluations
- alternating the learning side to prevent the agent from exploiting player-1-specific priority

[▶ Open Player 1 Priority Example Video](#)

Base Agent vs Base Agent

Video example for the Base Agent's preferred strategy

▶ Open Video

Main Limitation: Diversity

- Exploiters initialized from similar policies tend to discover similar weaknesses.
- Some strategically different exploits are hard to reach because they require first unlearning the Base Agent's preferred behavior.
- New initial agents with focuses on different unit types to improve exploiter diversity and Main Agent generalization.
- Hypothesis: LT therefore depends strongly on the diversity already present at initialization

Scope and Remaining Limitations

- evaluation is limited to one map: `basesWorkers16x16A`
- the setting uses full observability and no fog of war
- high computational cost
- exploiter diversity

Takeaway

The method works, but the strength of the final result depends strongly on the diversity and cost of the training setup.

Future Work

- train on multiple maps and with fog of war
- improve the diversity and strength of initial agents
- reduce computational cost of MicroRTS training
- investigate stronger diversity mechanisms for exploiters

Conclusion

Conclusion

LT with PPO improves the robustness and generalization of the MicroRTS Base Agent. Additional PPO fine-tuning improves the final agent even further.

Key takeaway

The effectiveness of LT depends strongly on exploiter diversity, which in turn depends on the diversity of the initial agents and BC experts.

- LT improves most matchups and the overall performance
- fine-tuning further improves the final agent
- diversity of initial agents is the critical practical factor

Thank you

Questions?

Main Agent vs League Exploiter

- League Exploiter resets
- different Hyperparameters
- Historical League Exploiter are not included as Main Exploiter opponents.

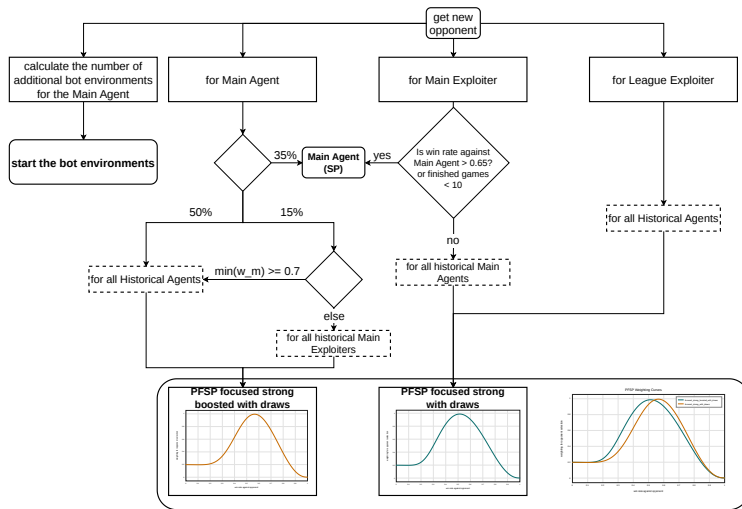
Why adjust the observations for SP?

- Gym-MicroRTS: Toward affordable full game real-time strategy games research with deep reinforcement learning implemented SP and concluded that a likely reason for worse performance of SP was that SP also has to train, to start as Player 2, while pure PPO only trained as Player 1.
- difficulties retrainig the Base Agent.

Why are the win rates against NativeMCTS low?

- it is one of the only opponents that give up winning the game and focus on staying far away from all opponents.
- This results in many draws and few losses.

Full LT Matchmaking



Matchmaking and opponent sampling in the LT setup.